# Ethan Yap

Year 2 Information Technology Student, Singapore Polytechnic

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## SUMMARY

IT student seeking a software engineering internship. Builds and deploys full-stack web apps in Node.js, Next.js and Python on PostgreSQL, from REST APIs to AI search, with Docker and GitHub Actions CI/CD.

## EDUCATION

**Singapore Polytechnic** · Diploma in Information Technology, Minor in Business — Apr 2025 – 2028 (expected)
**GPA 3.49 / 4.00** (Year 1) · Key modules: Back-End Development, Java Programming, Continuous Integration & Development, Software Engineering Principles, Design User Experience & Interface

**Maris Stella High School** · GCE O-Level, L1R5 net 10 · Direct School Admission (Robotics) — 2021 – 2024

## TECHNICAL SKILLS

| | |
|---|---|
| **Languages** | JavaScript, TypeScript, Python, Java, SQL, HTML, CSS |
| **Backend** | Node.js, Express, FastAPI, REST APIs, WebSockets, JWT authentication, background job queues |
| **Data & AI** | PostgreSQL, pgvector, MySQL, SQLite, pandas, ETL, embeddings, RAG, ChromaDB, Gemini API |
| **Frontend** | React, Next.js, Tailwind CSS, Bootstrap, Chart.js, Recharts, responsive and accessible UI, Java Swing |
| **DevOps & testing** | Docker, Docker Compose, nginx, GitHub Actions (CI/CD), Git, pytest, node:test, Zod, Figma |
| **Spoken** | English, Mandarin |

## PROJECTS

**Incident Monitoring Dashboard** · *Docker Compose, Node.js, PostgreSQL, WebSockets, Next.js* — Code · Demo
- Designed a 5-container Docker Compose stack with health checks, verified end to end by a smoke test in GitHub Actions.
- Built an alerting engine whose hysteresis (2 bad readings to open, 3 good to resolve) filters noisy spikes; 15 unit tests.

**AI Code Snippet & Documentation Hub** · *Next.js, TypeScript, PostgreSQL, pgvector, Gemini API* — Code · Demo
- Kept saves instant with a background worker on a Postgres SKIP LOCKED job queue that writes Zod-validated LLM docs.
- Built semantic search over 768-dimension embeddings with a pgvector HNSW index, re-ranked by Postgres full-text score.

**Academic Resource RAG Search Engine** · *Python, FastAPI, ChromaDB, sentence embeddings, pytest* — Code · Demo
- Built PDF ingestion that packs sentence-aligned chunks by page and section, so every answer cites its exact page.
- Refused off-topic questions with a relevance cut-off and rejected any uncited LLM answer, covered by 26 pytest tests in CI.

**Tech Trend Sentiment Tracker** · *Python, pandas, SQLite, FastAPI, Next.js, Recharts* — Code · Demo
- Automated a 6-hourly GitHub Actions ETL pipeline that de-duplicates news from 8 sources and scores its sentiment.
- Kept the Python API and the JavaScript dashboard in agreement with a parity test, one of 18 pytest tests run in CI.

**Pokémon Gym & Battle System** · *Node.js, Express, PostgreSQL, JWT, bcrypt* — Code · Demo
- Built a 45-endpoint REST API with JWT auth and ownership checks, then ported it from MySQL to PostgreSQL.
- Wrote a turn-based battle engine and a gacha shop in which every purchase runs as a single database transaction.

**MyLearningAI Study App** · *JavaScript, HTML, CSS, Figma (group of 4)* — Code · Demo
- Built our team's Figma design into a 60+ screen app; usability tests added a progress/time indicator and flag/skip.

**Student Internship Management System** · *Java, Swing, OOP (pair project)* — Code
- Built the enquiry system, multi-field search and per-company reports, then integrated it with my partner's admin system.

## WORK EXPERIENCE

**Two Men Bagel House** · Line Chef (part-time) — Feb 2025 – Apr 2026
- Coordinated orders between kitchen and front of house at peak service; ran the register and barista station.

**Bee Cheng Hiang** · Barbecue Cook (part-time) — Nov 2024 – Feb 2025
- Ran a cooking station alone through the festive peak, sequencing orders ahead of demand to shorten queues.

## LEADERSHIP, COMMUNITY & AWARDS

**Marist Ambassador** · frequent emcee at school events incl. prize-giving; volunteered at multiple Open Houses — 2021 – 2024

**Marist Robotics & Programming Club** · mentored juniors; Perse Coding Team Challenge 2022–23 (**Merit Award**) — 2021 – 2024

**Class Chairman** (Sec 4) · **Peer Supporter** (Sec 2) — 2022, 2024

**Community service** · ran registration for Mummy Yummy events of 100+ people; Magorium, Masjid Kassim — 2022 – 2025

**Edusave Character Award** · Ministry of Education and People's Association — 2024